

In-Person Lab - Assemble a Desktop Computer and Verify That it Works

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Before You Begin - How to Complete This Activity

Please refer to the [Lab Process Guide](#) for instructions on completing and submitting this activity.

Introduction

Disassembling a desktop computer is one thing; putting it back together is another. In this lab, you will assemble a desktop computer. You can use the one you disassembled in the previous lab, or for a more challenging experience, you can use one that was disassembled by a peer.

Safety first!

To ensure your safety and the safety of the electronic components, always verify that the computer is disconnected from any power source, and use the antistatic gear when disassembling it.

Note:

There are many different types of computer cases. In this lab, you will follow the steps for assembling a desktop computer in a generally logical manner. Still, during this lab, you may encounter situations where following the prescribed steps in sequential order is impossible. One hardware component may get in the way of installing another component. If this occurs, use your common sense and jump to the relevant step non-sequentially.

Objectives

At the end of this lab, you will be able to assemble a desktop computer and verify that it works.

Equipment/Requirements

A standard desktop computer (one per learner) in working condition that includes the following:

- Computer case.
- Power supply unit.
- CPU.
- Motherboard with live CMOS battery.

- Graphic card.
- RAM.
- Case fan.
- CPU cooler.
- HDD.
- SSD M.2.
- DVD (if available).
- PCI NIC.
- Internal cables.
- Monitor, keyboard, and mouse.
- Power cable.

In addition to the above list:

- Learner laptops with internet connection.
- Power outlet.
- A cellphone with a camera.
- Tech toolkit for each learner.

Assemble a desktop computer and verify that it works.

As an IT technician, you should be familiar with desktop computers. Replacing or upgrading computer hardware components is a common task.

Step 1: Prepare for desktop computer assembly.

- Set up your working environment. Ensure adequate lighting and antistatic protection.*
- Place the disassembled desktop computer case and components on the workbench.*
- Ensure that you have all desktop components, screws, and tools available.*

- d. *Use your smartphone to take a picture of the computer in its current state. Place the image in the box below.*



- e. **Step 2: Install the CPU.**

Before you start, Identify your computer's motherboard and ensure that you have the motherboard manual at hand. When assembling a computer, you will need this manual to help you with everything related to the assembly process.

- Locate the motherboard, remove it from the antistatic bag, and place it on the antistatic mat.*
- Search the internet for the **motherboard's manual** and have it handy. Paste the link to the manual into the box below.*

<https://www.dell.com/support/manuals/en-us/optiplex-9020-desktop/opt9020sffom-v2/specifications?guid=guid-0805b167-15c7-4302-a0b4-aabdc5e358a3&lang=en-us>

- c. *Locate the CPU, and remove it from its protective casing. DO NOT touch the conductive area or the pins on the bottom of the CPU.*
- d. *Clean the CPU with a thermal compound removal wipe if it has old thermal compound residue.*
- e. *Back on the motherboard, identify the location of the CPU socket.*
- f. *Identify the mechanism used to secure the CPU to the socket.*
Note: different CPU sockets have different mechanisms to secure the CPU. Regardless, all will have some type of lever you will need to release. Some sockets will have an additional cover plate you will need to open. All CPU sockets are ZIF, so do not use force. If you are unclear about how the CPU retaining mechanism works, consult the **motherboard manual**.
- g. *Release the CPU securing mechanism. No tools are necessary.*
- h. *Identify the CPU's orientation, which is usually marked on the motherboard and on the CPU. If you are experiencing difficulty understanding the proper CPU orientation, refer to the motherboard's manual.*
- i. *Once you know the orientation, gently place the CPU in the socket. You should place the CPU directly down without applying any force. The CPU should fit perfectly. If it does not fit properly, you probably have the orientation wrong. In that case, take the CPU out, verify the orientation, and try again. Take care not to touch the CPU or motherboard conductive areas or pins.*
- j. *Once the CPU is properly set in the socket, use the mechanism to close the socket and secure the CPU in place.*

- k. *Use your smartphone to take a picture of your motherboard with the CPU in the socket. Place the image in the box below.*

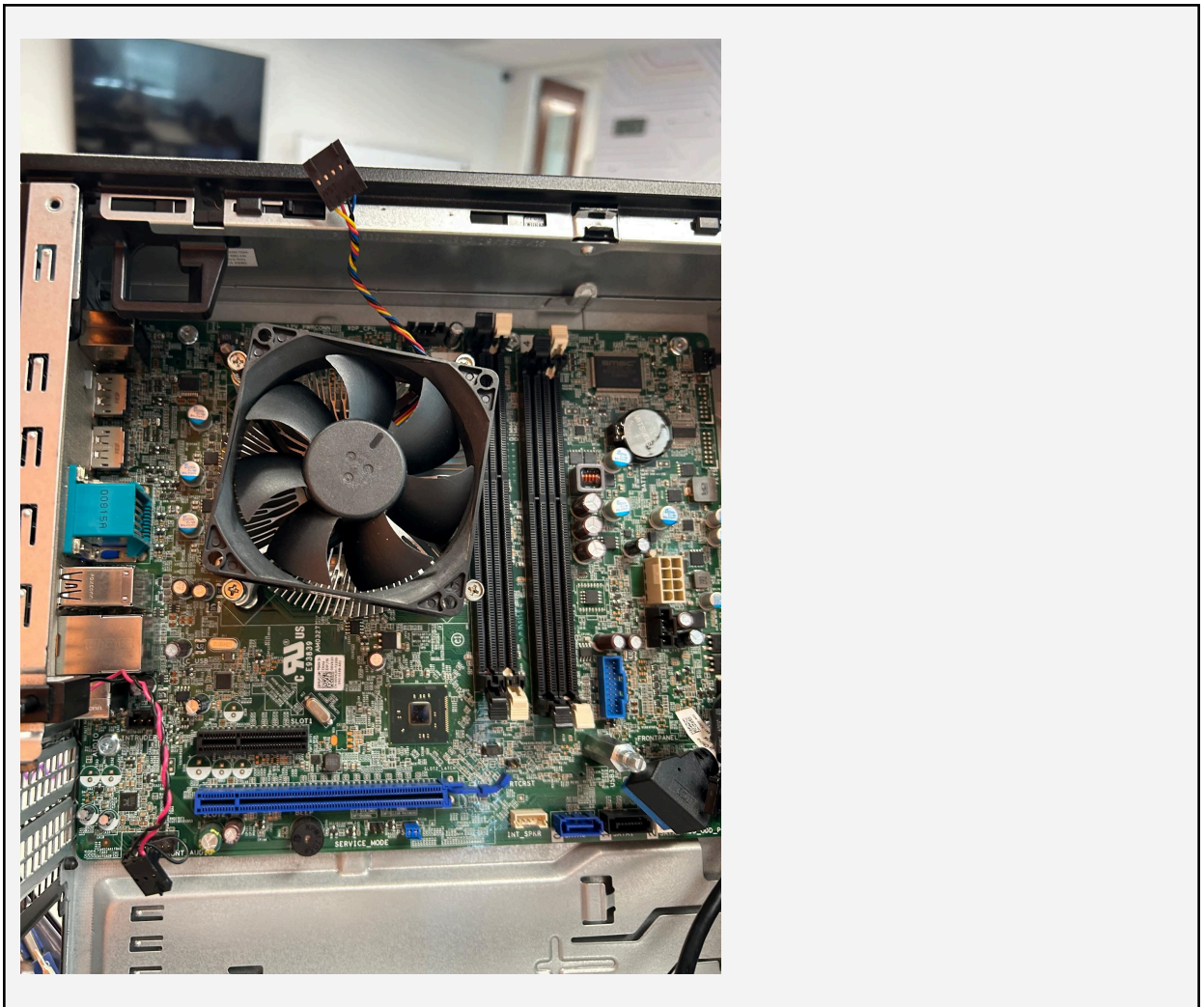


Step 3: Install the CPU cooling.

With CPU cooling being an extremely expansive world – there is just no end to cooling options for CPUs – installing the CPU cooling can be very easy or seriously challenging. If you run into problems with installing the CPU cooling, use the manual or conduct some research over the internet for how the cooler is installed.

- Retrieve the CPU cooler with all its related parts and screws, and place them on the workbench.*
- Use a thermal compound removal wipe to clean the cooler area that comes into contact with the CPU.*

- c. *Ensure that you understand how the CPU cooler is to be installed. Verify the cooler orientation, power cable and motherboard fan header, screws, and parts you are about to use.*
- d. *Back on the motherboard, place a small dab of thermal compound on the center of the CPU.*
- e. *Place the CPU cooler over the CPU and secure it into position. A properly mounted CPU cooler should be firmly attached to the motherboard.*
- f. *Connect the cooler power cable to the CPU fan header on the motherboard. Note that the connector is keyed and will only fit in one direction.*
- g. *Use your smartphone to take a picture of the installed CPU cooler. Place the image in the box below.*



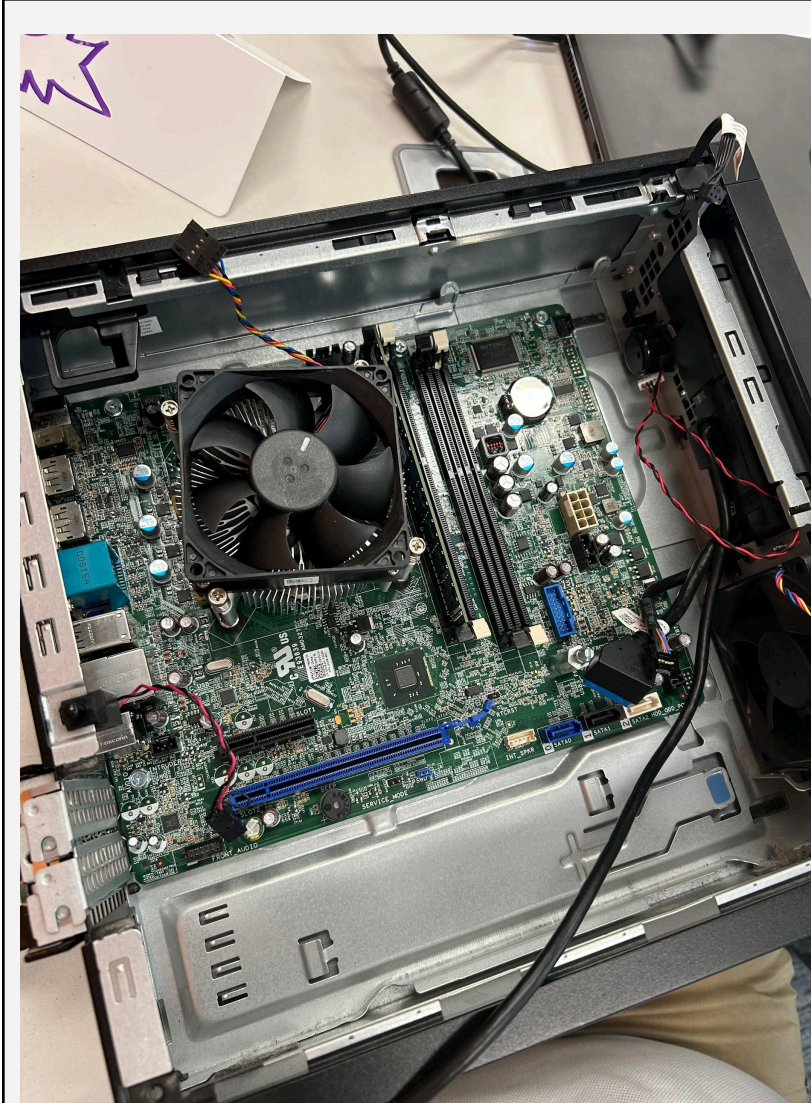
Step 4: Install the RAM.

- a. *Retrieve the RAM module(s) from the antistatic bags and lay them on the workbench. Do not touch the gold-plated conductors.*

- b. *How much memory will the computer have once you are finished assembling it? Type the answer into the box below.*

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- c. ***Refer to the motherboard manual*** to identify the correct slot(s) for RAM installation. The slot(s) to use may vary depending on the number of RAM modules and the motherboard's memory support options.
- d. *Release the clip(s) at the end of the RAM slot(s).*
- e. *Position the RAM module(s) over the slot(s). Use the central notch in every slot and DIMM to ensure the correct orientation for the module.*
- f. *With your fingers on both the RAM module ends, push the module directly downwards into the slot until the clip(s) snaps closed and secures the module in place. Note that this action may require you to use some force.*
- g. *Remove the power supply from the computer case and place it in your work area.*
- h. *Using your smartphone, take a photo of the motherboard with the installed RAM module(s) and place the image in the box below.*

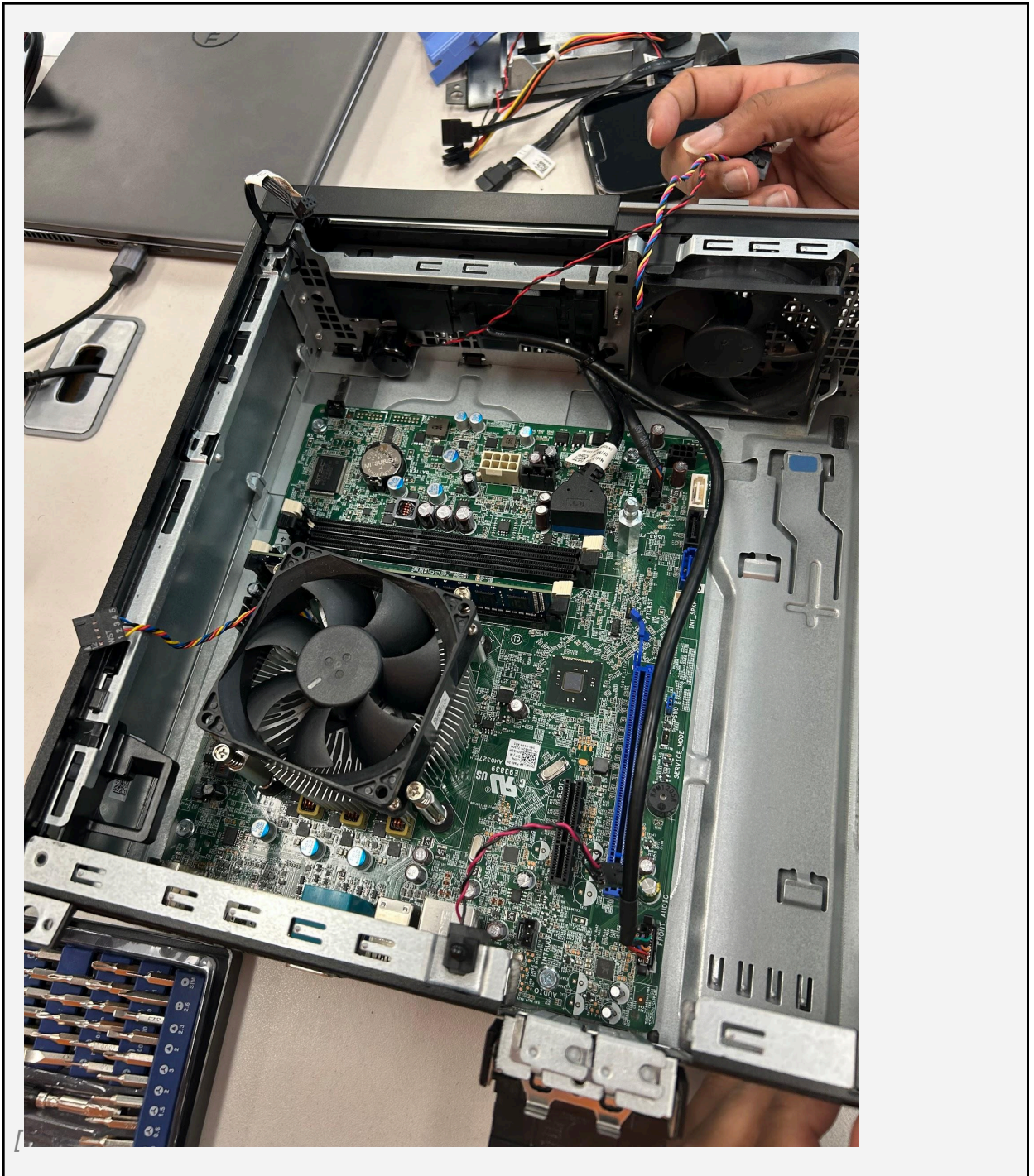


Step 5: Install the motherboard.

- Retrieve the computer case and position it on the workbench. Ensure that the case side panels are open and that the case's interior is accessible.*
- Retrieve the motherboard screws and place them on the workbench.*
- Inside the case, identify the location of the **standoffs** and **screws** used to secure the motherboard to the case.*
Note: some cases have the motherboard mounted to a detachable plate.
- Gently insert the motherboard into the case. Align the motherboard's I/O panel with the I/O plate in the case.*
- Align the motherboard's screw holes with the standoffs in the case.*

- f. *Use the motherboard's screws to secure the motherboard to the case. Note that there are usually 6 - 8 screws.*
- g. *Refer to the **motherboard's manual** to understand how to connect the case's front panel power button, LED indicator(s), front USB, and front audio to the motherboard. You can also reference the photo you took of these connectors during the computer disassembly lab.*
- h. *Once you understand how to connect the case's front panel power button, LED indicator(s), front USB, and front audio, run the cables in an **orderly manner** (use the cable management system if available) and **connect** the cables to the motherboard.*
- i. *If the answer is "**No**," move on to Step 7; otherwise, continue in this step.*
- j. *Identify the location of the **screws** securing the network adapter card to the case.*
Note: Not all cases have screws or require a screwdriver to remove extension cards; others may have a panel holding all cards in place.

- k. *Using your smartphone, take a photo of the case with the installed motherboard inside it and place the image in the box below.*



Step 6: Install the power supply unit.

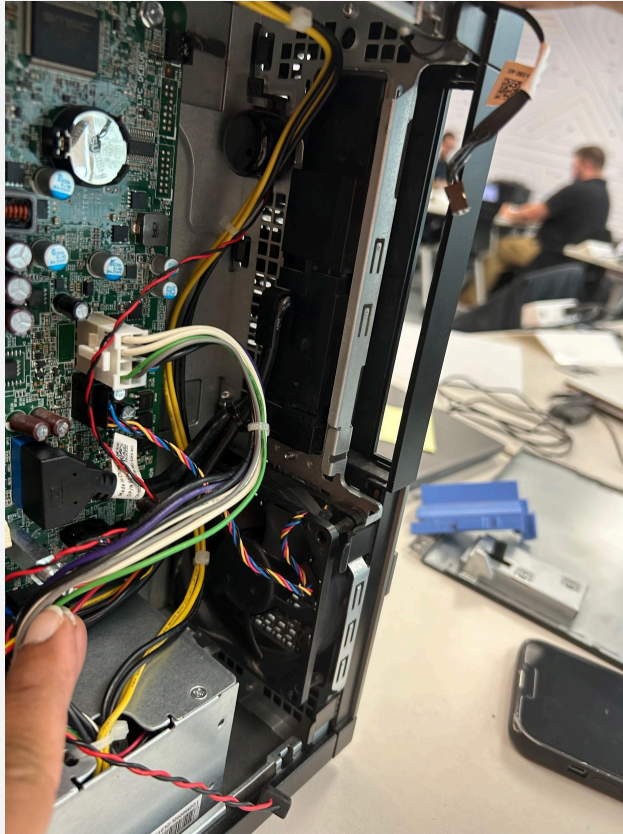
- a. *Retrieve the power supply unit and its screws and place them on the workbench.*

- b. *Identify the power supply's location in the case and the location of the screws used to secure the power supply unit to the case.*
Note: Not all cases have screws or require a screwdriver to remove power supplies; others may have a panel or cover holding the power supply in place. Use the case manual if necessary.
- c. *Set the power supply in its intended place inside the case and secure it in this position with the appropriate screws.*
- d. *Connect the motherboard to the power supply. Note that besides the main power connector, connecting auxiliary power connectors may be required. Refer to the motherboard's manual to ensure you perform the connections correctly.*
- e. *Using your smartphone, take a photo of the power supply mounted into the case and place the image in the box below.*



- f. *Step 7: Install the case fans.*
 - a. *Retrieve the case fan(s) and their screws, and place them on the workbench.*
 - b. *Explore the desktop case and identify the mounting location of all the case fans. Usually, case fans mounted on the back and top of the case push hot air out of the case, while fans mounted on the front and bottom pull cold air into the case.*
Note: The fan orientation will determine if the fan is pulling or pushing air, so ensure that you correctly orient the fan(s).
 - c. *Mount the case fan(s) to their original position. Secure the fan(s) in place with the appropriate fan screws.*
 - d. *Connect the fan(s) power cables to the fan power headers on the motherboard. Ensure that you run the cables in an orderly manner. Use a cable management system if one is available in this case.*

- e. *Use your smartphone to take a picture of the case with the fan(s) mounted. Place the image in the box below.*



Step 8: Install the storage device(s).

In this step, you will install all internal storage devices into the computer. These may include one or more HDDs, SSDs, and optical drives.

Note that different computer cases offer different bays to accommodate storage devices, and bays may be located in different locations. In addition, storage devices using interfaces such as M.2 may be connected directly to the motherboard and hidden under a cooling plate.

Regardless of their number, install all of the storage devices available into the case or directly onto the motherboard (M.2).

- Retrieve the storage device(s) and their screws and lay them on the workbench.*
- Explore the inside of the case and identify the storage device(s) mounting location and the mechanism used to mount them.*
- Mount the storage device(s) to its original position. Secure the device(s) in place with the appropriate screws.*

- d. *Connect the storage device(s)'s power to the appropriate power cables coming from the power supply (an M.2 storage device does not require a power cable). Be sure to run the cables in an orderly manner. Use a cable management system if one is available in this case.*
- e. *Connect the storage device(s)'s data connectors to the appropriate data cables. Connect the other end of the data cable(s) to the correct connector on the motherboard (an M.2 storage device does not require a data cable). Be sure to run the cables in an orderly manner. Use a cable management system if one is available in this case.*

- f. *Use your smartphone to take a picture of the storage device(s) mounted and connected inside the case. Place the image in the box below.*



Step 9: Install the graphic adapter card.

- a. Do you have a graphic adapter card that you can install into an expansion slot? Type your answer into the box below.

No

- b. If you answered “**No**,” move on to the next step; otherwise, continue in this step.
- c. Retrieve the graphic adapter card and the card mounting screws and lay them on the workbench.
Note: Not all cases have screws or require a screwdriver to mount extension cards; others may have a panel to hold cards in place.
- d. Identify the motherboard slot that will hold the graphic card.
- e. Identify the location of the screw(s) or retaining mechanism on the case for securing the card in place.
- f. Drop the graphic card into the slot. Ensure the clip at the slot’s end is locking the card to the slot.
- g. Secure the graphic card to the case with screws or with the securing mechanism specific to your case.
- h. Consult the **graphic’s card manual** and, if necessary, connect the power cable(s) from the power supply to the correct connector on the card.
- i. Using your smartphone, take a photo of the mounted graphic card and place the image in the box below.

N/A

Step 10: Install the network adapter card.

- a. Do you have a network adapter card that you can install into an expansion slot? Type your answer into the box below.

N/A

- b. If you answered “**No**,” move on to the next step; otherwise, continue in this step.

- c. Retrieve the network adapter card and the card mounting screws and lay them on the workbench.
Note: Not all cases have screws or require a screwdriver to mount extension cards; others may have a panel to hold cards in place.
- d. Identify the motherboard slot that will hold the network card.
- e. Identify the location of the screw(s) or retaining mechanism on the case for securing the card in place.
- f. Drop the network card into the slot. If applicable, ensure the clip at the slot's end locks the card to the slot.
- g. Secure the network card to the case with screws or with the securing mechanism specific to your case.
- h. Using your smartphone, take a photo of the mounted network adapter card and place the image in the box below.

N/A

Step 11: Install the additional extension card.

- a. Do you have an additional extension card that you can install into an expansion slot?
Type your answer into the box below.

N/A

- b. If you answered "**No**," move on to the next step; otherwise, continue in this step.
- c. Retrieve the additional extension and the card mounting screws and lay them on the workbench.
Note: Not all cases have screws or require a screwdriver to mount extension cards; others may have a panel to hold cards in place.
- d. Identify the motherboard slot that will hold the extension card.
- e. Drop the expansion card into the slot. If applicable, ensure the clip at the slot's end locks the card to the slot.
- f. Secure the expansion card to the case with screws or with the securing mechanism specific to your case.

- g. *Using your smartphone, take a photo of the mounted expansion adapter card and place the image in the box below.*

N/A

Step 12: Inspect the case.

- a. *Carefully examine the inside of the case.*
- b. *Ensure that all hardware is sitting properly and securely in the intended locations.*
- c. *Ensure that all internal cables are correctly connected and run in an orderly manner.*
- d. *When you feel confident that everything is in order, use your smartphone to take a picture of the inside of the computer case and place the image in the box below.*



Step 13: Verify the computer works.

- a. *With the case still open, connect the monitor, keyboard, and mouse and plug the computer into a power outlet.*
- b. *Power on the computer.*
- c. *If your computer is having issues with powering up, apply a troubleshooting methodology to resolve the issue(s).*
- d. *Did the computer display the BIOS splash screen and BIOS access keys sequence? Type your answer into the box below.*

Yes

- e. *Did the computer load an operating system? Type your answer into the box below.*

Yes

- f. *Use your smartphone to take a picture of the computer and its monitor while powered on. Place the image in the box below.*

it Powered on

Step 14: Close the case.

- a. *Retrieve the case side panels and screws and place them on your workbench.*
 - b. *Reassemble the case side panels.*
 - c. *Identify the location of the screws holding the side panels to the case frame.*
Note: Not all cases have screws or require a screwdriver to secure the side panels.
 - d. *Secure the screws holding the side panels to the case. Use a screwdriver if necessary.*
 - e. *Using your smartphone, take a photo of your closed computer case and place the image in the box below.*
- Step 15: Store the computer and clear the workbench.

We hope you enjoyed this lab. The time to clean up has come.

- a. *Return the desktop computer to the storage area.*
- b. *Gather all the tools and antistatic gear and move them to the storage area.*

- c. *Ensure that your work area is tidy and clean.*